

WESTERN GHATS SENTINEL

Tree Cover • Carbon • Conservation

Environmental Intelligence & Data Analytics Report

Study Region: Karnataka • Kerala • Tamil Nadu • Goa • Maharashtra

Analysis Period: 2001 – 2025

Analytical Focus: Forest loss • Carbon • Spatial intensity • Conservation

Prepared as a data analytics and environmental intelligence project.

1. Executive Summary

Western Ghats Sentinel is an interactive environmental analytics project designed to examine tree-cover loss, carbon-related indicators, spatial patterns and conservation information across five states associated with the Western Ghats: Karnataka, Kerala, Tamil Nadu, Goa and Maharashtra.

The available annual dataset covers **2001–2025** and records approximately **237,864 hectares** of tree-cover loss. The average annual loss is approximately **9,515 hectares**.

Indicator	Result
Total tree-cover loss	237,864 ha
Average annual loss	9,515 ha
Peak loss year	2023
Peak annual loss	21,159 ha
Highest total-loss state	Kerala
Highest loss-intensity state	Kerala
Highest emissions state	Kerala
Species represented	27

2. Project Objective

The project transforms environmental datasets into an interactive analytical system that allows users to investigate forest-loss patterns, compare states, understand normalized spatial intensity, examine carbon-related indicators and explore conservation-status information.

Primary objectives

- Measure annual tree-cover loss across the available study period.
- Compare tree-cover loss between the five study states.
- Normalize loss using state-level area to provide spatial intensity.
- Examine carbon and CO₂-related indicators associated with forest loss.
- Explore temporal conservation-status information.
- Investigate statistical relationships and lagged associations.
- Present the results through an interactive Streamlit dashboard.

3. Research Questions

- How has tree-cover loss changed over the analysis period?
- How does forest loss differ among the five study states?

- Which states show greater loss intensity after area normalization?
- How are tree-cover loss and carbon-related indicators associated?
- What temporal patterns are visible in conservation-status data?
- Are there observable lagged relationships between environmental indicators?
- What can the available datasets reveal, and what are their analytical boundaries?

4. Study Region

The geographical focus of this project is the Western Ghats study region represented through five Indian states: **Karnataka, Kerala, Tamil Nadu, Goa and Maharashtra**.

The spatial component uses polygon boundaries representing the study states. District-level boundary information is also available in the underlying spatial data; however, the tree-cover-loss observations used for the principal state-level analysis are aggregated at the state level.

Therefore, state-level results should not be interpreted as district-level tree-cover-loss estimates.

5. Data & Dataset Architecture

Dataset	Analytical purpose
FOREST_RAW_DATA.csv	Underlying forest/environmental observations
ANNUAL_TREE_COVER_LOSS.csv	Annual tree-cover-loss analysis
STATE_YEAR_TREE_LOSS.csv	State-by-year forest-loss analysis
FINAL_STATE_ANALYSIS.csv	State-level environmental indicators
SPECIES_STATUS_HISTORY.csv	Historical species conservation-status records
SPECIES_STATUS_TRENDS.csv	Species-status trend analysis
SPECIES_STATUS_TRANSITIONS.csv	Conservation-status transition analysis
LAG_ANALYSIS.csv	Lagged relationship analysis
map_complete.geojson	Spatial boundary and mapping layer

6. Data Preparation & Validation

Data preparation involved loading the source datasets, standardizing analytical fields, aggregating state-level observations and preparing spatial information for visualization.

Spatial validation

A key spatial-validation step was required because the original geographic data contained multiple district polygons while the tree-cover-loss values were available at state level. Directly joining state-level totals to district geometries would repeat the same state value across multiple polygons.

To address this, district geometries were dissolved into a single polygon per state before the state-level environmental totals were joined. This prevents duplicated state totals in the mapped analysis.

Area calculations use an equal-area coordinate reference system so that the normalized loss-intensity indicator is based on projected area rather than unprojected geographic coordinates.

7. Spatial Methodology

- State geometries are dissolved before state-level totals are joined.
- Area is calculated using an equal-area coordinate reference system.
- Loss intensity is expressed as tree-cover loss per 1,000 km².
- Carbon-related indicators are displayed alongside forest-loss measures.
- The spatial explorer provides state-level visual comparison.

8. Forest-Loss Analysis

Across the available annual data, the project records approximately **237,864 hectares** of tree-cover loss during the analysis period. The average annual loss is approximately **9,515 hectares**.

The highest annual loss occurs in **2023**, with approximately **21,159 hectares** recorded for that year.

At the state level, **Kerala** records the highest total tree-cover loss in the available state-level analysis, at approximately **117,145 hectares**.

9. Carbon & Emissions Analysis

The state-level dataset includes estimated emissions indicators associated with the environmental analysis. Within that dataset, **Kerala** records the highest total emissions value, approximately **74,623,198 Mg**.

The available annual data produces a Spearman correlation of approximately **0.992** between tree-cover loss and the identified emissions indicator.

The corresponding Pearson correlation is approximately **0.995**.

Correlation describes statistical association. It does not by itself establish that one environmental variable causes another.

10. Conservation Analysis

The conservation component contains historical species-status information representing approximately **27 unique species** across **7 status categories** in the available history dataset.

The dashboard examines status distributions, historical records, conservation trajectories and status transitions where supported by the available datasets.

These records provide temporal conservation context but do not establish that forest loss caused a particular species-status change. Species geographic-range information is not included in the analytical dataset, so direct habitat-loss attribution cannot be established from these records alone.

11. Relationship & Lag Analysis

- Annual tree-cover loss and emissions are compared statistically where compatible fields are available.
- Pearson correlation measures linear association.
- Spearman correlation measures monotonic association.
- State-level relationships provide a cross-sectional comparison.
- Lag analysis examines relationships at different temporal offsets.
- Statistical association is not treated as proof of causation.

12. Key Findings

Forest-loss scale	The available dataset records approximately 237,864 hectares of tree-cover loss.
Peak year	The highest annual loss occurs in 2023, with approximately 21,159 hectares.
State comparison	Kerala records the highest total tree-cover loss among the five study states.
Spatial intensity	Kerala records the highest normalized loss intensity in the state-level dataset.
Carbon pressure	Kerala records the highest total emissions indicator in the state-level dataset.
Conservation context	The conservation history dataset represents approximately 27 unique species across 7 status categories.

13. Application Architecture

Layer	Technology / implementation
Application interface	Streamlit
Data processing	Python + Pandas
Spatial analysis	GeoPandas
Interactive visualization	Plotly
Spatial data	GeoJSON
Statistical analysis	Pandas / numerical analysis
Report generation	ReportLab
Data storage	CSV + GeoJSON

14. Technology Stack

- Python
- Streamlit

- Pandas
- GeoPandas
- Plotly
- ReportLab
- CSV
- GeoJSON
- HTML/CSS styling

15. Research Limitations

- Tree-cover-loss observations are represented at state level.
- State-level results should not be interpreted as district-level estimates.
- Correlation does not establish causation.
- Lagged statistical relationships do not independently establish causal mechanisms.
- The conservation dataset does not contain species geographic-range information.
- Species-status associations therefore cannot be treated as direct habitat-loss attribution.
- Environmental indicators may represent different measurement concepts and should be interpreted according to their source definitions.
- The dashboard is an analytical representation of the available datasets and is not a complete ecological assessment of the Western Ghats.

16. Future Scope

- Integrate additional satellite-derived environmental indicators.
- Add district-level forest-loss data where compatible datasets are available.
- Integrate species geographic-range or habitat-distribution data.
- Add protected-area and biodiversity-hotspot boundaries.
- Introduce more advanced spatial statistics.
- Develop automated data-refresh pipelines.
- Add scenario modelling and forecasting where scientifically appropriate.
- Deploy the dashboard publicly through Streamlit Community Cloud.
- Enable user-selectable report generation directly from the dashboard.

17. Conclusion

Western Ghats Sentinel demonstrates how environmental datasets can be transformed into an interactive data-storytelling and analytical system. The project combines forest-loss analysis, spatial comparison, carbon-related indicators, conservation information and statistical relationships within a single application.

Rather than relying on a single environmental indicator, the project combines multiple measures to provide broader analytical context. The methodology also distinguishes observed patterns and statistical associations from causal conclusions.

The resulting system provides a foundation for further environmental analytics, spatial research and data-driven conservation investigation.

